

EEP2 – Week 2

24/1/2025 – E4A-06-03

A0287717M – Brians Tjipto Meidianto

1)

Frequency: 1.0000

1 (a)

Amplitude: 432mV

Peak-to-Peak Amplitude: 864mV

1 (b)



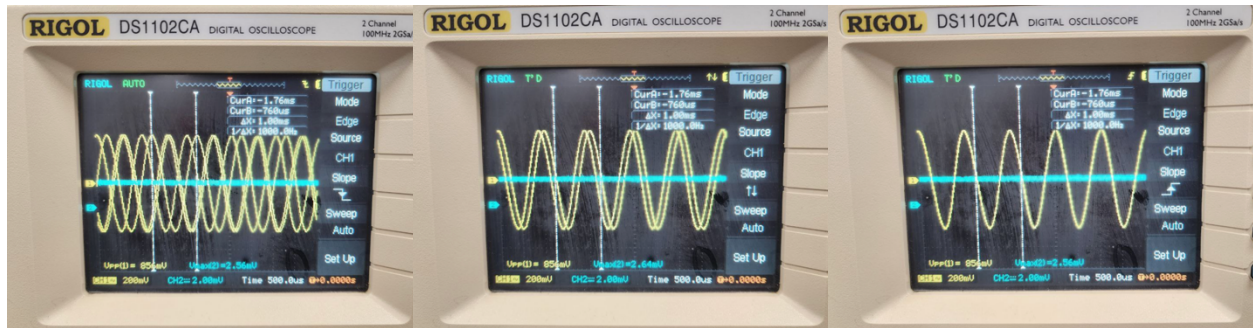
- i) Offset in DC coupling changes the midpoint
- ii) Offset in AC coupling warps the peaks

1 (c)

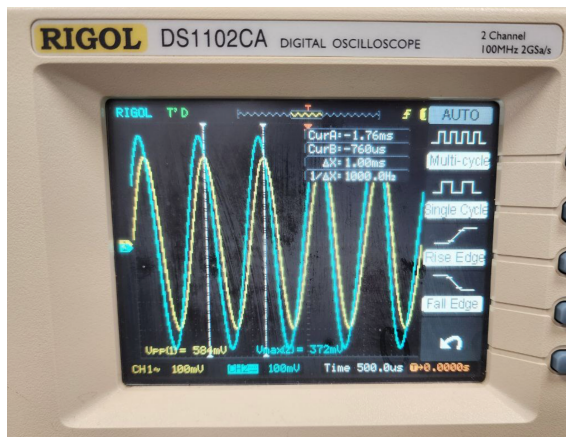
Period: 1ms

Frequency: $1/1\text{ms} = 1\text{kHz}$

1 (d)



1 (e)



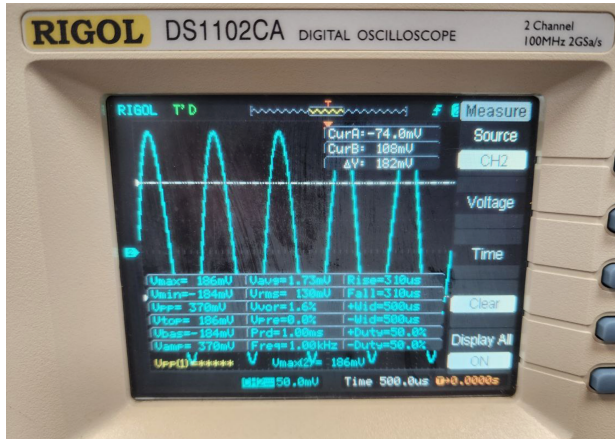
Period T second: 1ms

Offset ΔT second: 100us

$$\phi = \frac{\Delta T}{T} \times 2\pi \text{ rad/s}$$

$$\phi = \frac{0.001}{1} \times 2\pi = 0.00628 \text{ rad/s}$$

1 (f)



Amplitude of current: $370\text{mV}/2 = 185\text{mV}/100 = 0.00185\text{A} = 1.85\text{mA}$

Phase-shift w.r.t. the input voltage: 0.00628 rad/s